Equilibrium: find the missing balance

LEARNING TARGET I can find a missing force when acceleration is zero.

SECTION A KEY VOCABULARY

Equilibrium	Acceleration is zero. The forces sum to zero in every direction.
Two kinds	At rest *or* moving at constant velocity.
Missing force	Whatever is needed to make each direction balance.
Careful	Zero velocity for an instant does not guarantee equilibrium.

SECTION B KEY CONDITION

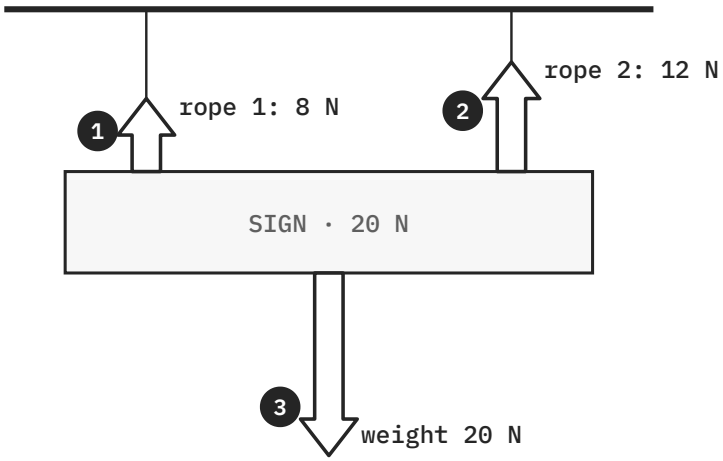
$$\Sigma F = 0$$

up = down
right = left

Σ "sum of"
force newtons (N)

SECTION C COLOR IT · SUPPORTS MUST TOTAL THE WEIGHT

8 N + 12 N = 20 N.



☐ **1 · ORANGE** Rope 1 tension
☐ **2 · ORANGE** Rope 2 tension
☐ **3 · ORANGE** Weight (W)

SECTION D WORKED EXAMPLE

A 40 N sign hangs at rest from two vertical ropes. One tension is 15 N. Find the other.

GIVEN	EQUATION	WORK	ANSWER
weight 40 N rope 1 = 15 N at rest	up = down	$15 + T_2 = 40$ $T_2 = 40 - 15$	**25 N up**

SECTION E THE 2026-XJ THREAD

DAY 18 OF 22 · NEW TOOL: EQUILIBRIUM

If the forces on XJ balanced, its acceleration would be zero and its velocity would stay constant. Analysts check for balance before assuming steady motion.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else